

TOPIC

SUMMARY LESSON

UNIT

1

1. Turning effect of force

UNIT

2

- 1. Lever**
- 2. Inclined Plane**
- 3. Gear and Belt System**
- 4. Pulley Wheel and Axle**

UNIT

3

- 1. Electrical Energy**
- 2. Household Wiring System**

TOPIC

SUMMARY LESSON

UNIT

4

- 1. The magnetic field**
- 2. Magnetic Force**

UNIT

5

- 1. Light**
- 2. Reflection**
- 3. Mirror**
- 4. Refraction**
- 5. Lenses**

Prepared By:

Instructor: MOEUN Sokmean



- 1 Turning effect of a force**
- 2 Simple Machines**
- 3 Electricity**
- 4 Magnetic Field**
- 5 Light**
- 6 Summary**



Chapter1:

Turning effect of force



Turning effect of force

Resultant Force

- Opposite direction

$$F = |a - b|$$

- Same direction

$$R = a + b$$

- At 90° angle

$$R = \sqrt{a^2 + b^2}$$

- At angle α

$$R = \sqrt{a^2 + b^2 + 2ab \cos \alpha}$$

Resolving Force

- Horizontal Force

$$F_x = F \cos \theta$$

- Horizontal Force

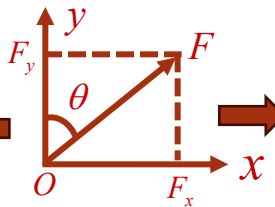
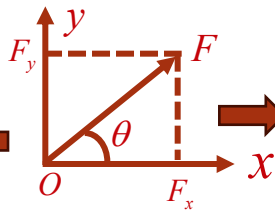
$$F_x = F \sin \theta$$

- Direction of a resultant force

$$\theta = \tan^{-1} \left(\frac{\text{opp}}{\text{adj}} \right) = \tan^{-1} \left(\frac{b}{a} \right)$$

Opposite side: b (m or N)

Adjacent side: a (m or N)



- Vertical Force

$$F_y = F \sin \theta$$

- Vertical Force

$$F_y = F \cos \theta$$

Moment

Moment of a force

$$M = F \times d_{\perp}$$

Net moment

$$M_{net} = M_{ACW} - M_{CW}$$

Moment of a force: M (Nm)

Force: F (N)

Perpendicular distance: $d_{\perp}$ (m)

Two parallel forces

Principle of moment: $M_{ACW} = M_{CW}$

$$F_1 d_1 = F_2 d_2 \Rightarrow \frac{F_1}{F_2} = \frac{d_2}{d_1}$$

Force F_A : $F_A = \frac{F \times OB}{AB}$

Force F_B : $F_B = \frac{F \times OA}{AB}$





Simple Machine

Lever

Inclined Plane

Pulley

Gears System

Mechanical advantage

$$MA = \frac{F_R}{F_E} = \frac{d_E}{d_R}$$

Efficiency: eff

$$eff = \frac{F_R \times d_R}{F_E \times d_E} \times 100\% = \frac{MA}{VR} \times 100\%$$

Load: F_R (N)

Efford: F_E (N)

Load arm length: d_R (m)

Efford arm length: d_E (m)

Mechanical advantage

$$MA = \frac{F_R}{F_E} = \frac{L}{h} \left\{ \begin{array}{l} \text{Ramp: } L \text{ (m)} \\ \text{Height: } h \text{ (m)} \end{array} \right.$$

Work done by load and effort

$$W_E = F_E \times L, W_R = F_R \times h$$

If no friction:

$$W_E = W_R \Leftrightarrow F_E L = F_R h$$

Efficiency:

$$eff = \frac{F_R \times d_R}{F_E \times d_E} \times 100\%$$

Mechanical advantage

$$MA = \frac{F_R}{F_E} = \# \text{ of supporting cables}$$

Velocity Ratio

$$VR = \frac{d_E}{d_R} \left\{ \begin{array}{l} \text{Pulling distance: } d_E \text{ (m)} \\ \text{Lifting distance: } d_R \text{ (m)} \end{array} \right.$$

Efficiency:

$$eff = \frac{F_R \times d_R}{F_E \times d_E} \times 100\% = \frac{MA}{VR} \times 100\%$$

Mechanical advantage

$$MA = \frac{F_R}{F_E} = \frac{N_R}{N_E} = \frac{r_R}{r_E} = \frac{d_R}{d_E}$$

Gear Ratio (GR):

$$GR = \frac{N_R}{N_E} = \frac{d_R}{d_E}$$

Relation of turns-teeth

$$N_E t_E = N_R t_R \Leftrightarrow \frac{N_E}{N_R} = \frac{t_R}{t_E}$$

Efficiency:

$$eff = \frac{F_R \times d_R}{F_E \times d_E} \times 100\%$$





Belt System

Simple Machine

Wheel and Axle

Mechanical advantage

$$MA = \frac{F_R}{F_E} = \frac{R_R}{R_E}$$

Radius of driver wheel: R_E (m)
Radius of driven wheel: R_R (m)

Turns and diameter relation

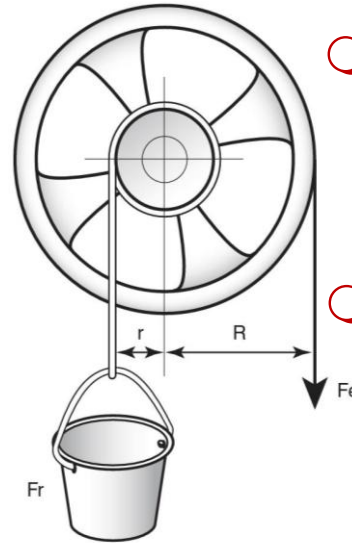
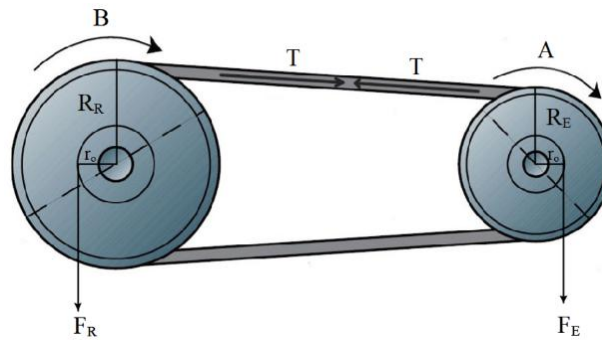
$$N_E D_E = N_R D_R$$

Turns of driver wheel: N_E

Turns of driven wheel: N_R

Diameter of driver wheel: D_E (m)

Diameter of driven wheel: D_R (m)



Moment of effort and load

$$M_E = F_E \times R, M_R = F_R \times r$$

Mechanical advantage

$$MA = \frac{F_R}{F_E} = \frac{R}{r}$$

Radius of a wheel: R (m)

Radius of an axle: r (m)

Velocity ratio: VR

Efficiency: eff

Velocity Ratio:

$$VR = \frac{\text{distance moved by effort}}{\text{distance moved by load}} = \frac{2\pi R}{2\pi r} = \frac{R}{r}$$

Efficiency:

$$eff = \frac{F_R \times r}{F_E \times R} \times 100\% = \frac{MA}{VR} \times 100\%$$





Electrical Power

Electricity

Heat Energy

Definition of power

$$P = \frac{W}{t} = \frac{E}{t}$$

Power of current

$$P = VI = RI^2 = \frac{V^2}{R}$$

Energy Transfer

$$E = QV = VIt = RI^2t = \frac{V^2}{R}t$$

Work done: W (J)
 Energy transfer: E (J)
 Electrical power: P (W)
 Time taken: t (s)

Voltage: V (V)
 Resistance: R (Ω)
 Electric current: I (A)
 Electric charge: Q (C)

Heat energy: Q (J)

Mass of a substance: m (kg)

Specific heat capacity: c (J/kg. $^{\circ}$ C)

Temperature change: ΔT ($^{\circ}$ C)

Cost of using electricity

$$\text{cost} = E(\text{kWh}) \times \text{price}/1\text{kWh}$$

Heat gained or lost

$$Q = mc\Delta T$$

Specific heat capacity

$$c = \frac{Q}{m\Delta T}$$

Conservation of energy:

$$E = Q \Leftrightarrow RI^2t = mc\Delta T$$

Total Energy:

$$E_L = E \times \% \text{ of light}$$

Light Energy: E_L (J)

$$E_H = E \times \% \text{ of heat}$$

Heat Energy: E_H (J)



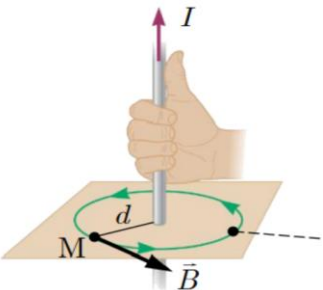


Magnetic Field

- Due to a straight wire

$$B = \frac{\mu_0 I}{2\pi d}$$

- Direction

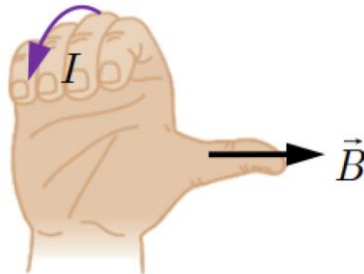


Magnetic field lines

Permeability of free space: $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$
 Electric current: $I \text{ (A)}$
 Distance from a wire: $d \text{ (m)}$
 Number of turns: $N \text{ (turns)}$
 Length of a solenoid: $l \text{ (m)}$

- Due to a solenoid

$$B = \frac{\mu_0 NI}{l}$$



- Direction

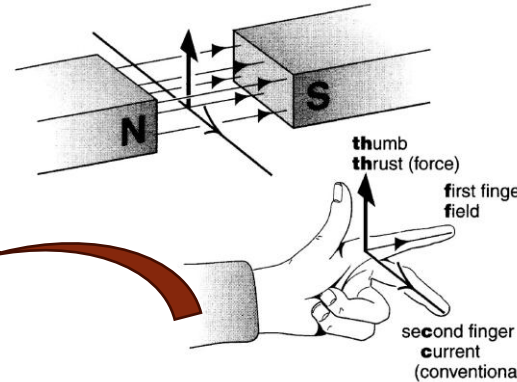
Magnetism

Magnetic Force

- On a current carrying wire

$$F = IlB \sin \theta$$

- Direction of magnetic force



Magnetic force: $F \text{ (N)}$
 Magnetic field: $B \text{ (T)}$
 Length of a wire: $l \text{ (m)}$

Transformers

- Voltage, Turn and Current Ratio

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S}$$

$$\frac{N_S}{N_P} > 1 : \text{Step-up}$$

$$\frac{N_S}{N_P} < 1 : \text{Step-down}$$

- Power loss:

$$P_J = RI^2, P_J = P_{e1} - P_{e2}$$

$$P_{e1} = V_P I_P$$

$$P_{e2} = V_S I_S$$

Secondary current: $I_S \text{ (A)}$
 Primary current: $I_P \text{ (A)}$
 Secondary voltage: $V_S \text{ (V)}$
 Primary voltage: $V_P \text{ (V)}$
 Secondary turns: $N_S \text{ (turns)}$
 Primary turns: $N_P \text{ (turns)}$





Reflection of light

Light

Refraction of light

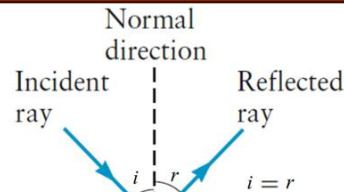
- The Law of Reflection states that:

$$i = r$$

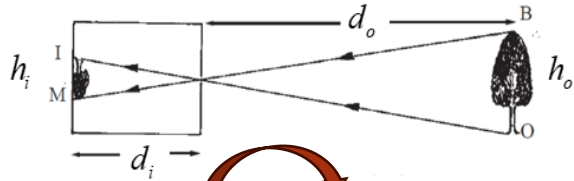
- Pinhole Camera

$$m = \frac{h_i}{h_o} = \frac{d_i}{d_o}$$

Image's height: h_i (m)
 Object's height: h_o (m)
 Image's distance: d_i (m)
 Object's distance: d_o (m)



Angle of incidence: i ($^\circ$)
 Angle of reflection: r ($^\circ$)



- Shadows

Umbra
 Central dark region
 Penumbra
 Partial shadow

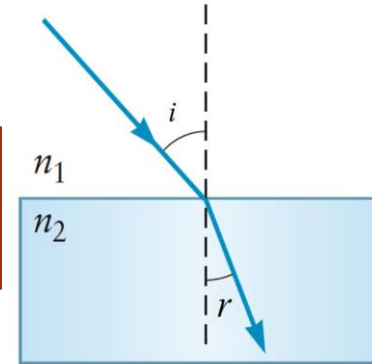
- Relationship between real depth and apparent depth

$$n = \frac{\text{real depth}}{\text{apparent depth}} = \frac{H}{h}$$

- Refractive index

$$n = \frac{c}{v}$$

Real depth: H (m)
 Apparent depth: d (m)
 Speed of light in air: $c = 3 \times 10^8$ m/s
 Speed of light in a medium: v (m/s)
 Refractive index: n (unitless)



- Snell's Law

$$n_1 \sin i = n_2 \sin r$$

- Critical Angle

$$\sin \theta_c = \frac{n_2}{n_1}$$

Critical angle: θ_c ($^\circ$)
 First index: n_1
 Second index: n_2
 Incidence angle: i
 Refractive angle: r





Plane Mirrors

Mirrors

Curve Mirrors

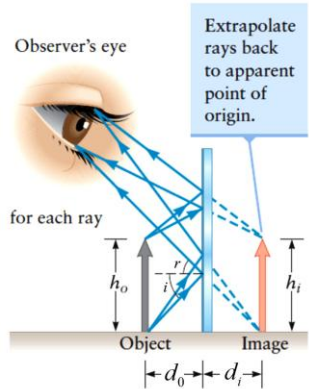
- The Law of Reflection states that:

$$i = r$$

Angle of incidence: i ($^\circ$)

Angle of reflection: r ($^\circ$)

- Characteristic of the image



Image's distance: d_i (m)

Object's distance: d_o (m)

- Mirror's Equation

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow f = \frac{d_i d_o}{d_i + d_o}$$

- Object's distance

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow d_i = \frac{f d_o}{d_o - f}$$

- Image's distance

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow d_o = \frac{f d_i}{d_i - f}$$

- Radius & focal length relation

$$f = \frac{R}{2} \Rightarrow R = 2f$$

- Magnification of mirror

$$M = \frac{d_i}{d_o} = -\frac{h_i}{h_o}, h_i = -\frac{d_i h_o}{d_o}$$

Focal length: f (cm, m)

Image's height: h_i (cm, m)

Object's height: h_o (cm, m)

Radius of curvature: R (cm, m)





Lenses

○ Lenses' Equation

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow f = \frac{d_i d_o}{d_i + d_o}$$

○ Object's distance

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow d_i = \frac{f d_o}{d_o - f}$$

○ Image's distance

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \Rightarrow d_o = \frac{f d_i}{d_i - f}$$

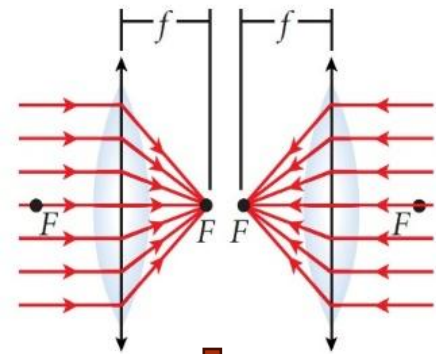
Convex lenses: $f > 0$ Concave lenses: $f < 0$ Focal length: f (cm, m)Image's height: h_i (cm, m)Object's height: h_o (cm, m)Radius of curvature: R (cm, m)Image's distance: d_i (cm, m)Object's distance: d_o (m, cm)

○ Radius & focal length relation ○ Convex lens

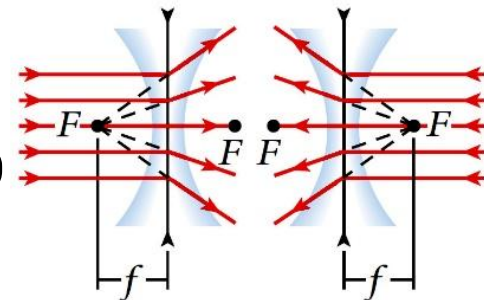
$$f = \frac{R}{2} \Rightarrow R = 2f$$

○ Magnification of mirror

$$M = \frac{d_i}{d_o} = -\frac{h_i}{h_o}, h_i = -\frac{d_i h_o}{d_o}$$

Image is upright: $h_i > 0$ Image is inverted: $h_i < 0$ Magnification: M 

○ Concave lens



Q & A

Thank You!